

Measurement-Based Multipath Characterization of GPS L1 C/A Signals for Vehicular Satellite-to-Ground Links in Challenging Environments

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Abstract— Vehicular satellite positioning is challenged by multipath and rapidly changing satellite-to-receiver geometry along dense urban and mountain/valley roads. As terrestrial and non-terrestrial systems become increasingly integrated for 6G localization, measurement-based characterization of mobile satellite-to-ground propagation remains important. This paper extracts and statistically characterizes GPS L1 C/A multipath components (MPCs) from moving-vehicle measurements. Navigation-data alignment forms coherent 40-ms observations, SAGE estimates delay, Doppler, and complex gain, and temporal-consistency screening retains persistent MPCs. The analysis uses 518 observations from 236 satellite tracks. Environment-specific models describe excess delay, signed relative Doppler, relative power, and the power–delay relation, with complete-scene complexity checks supporting the reported Doppler and power representations. These path-level descriptions provide inputs for simplified vehicular satellite-to-ground simulation and are relevant to GNSS and NTN-assisted localization under challenging propagation conditions.

1. Core method

Measurement and signal processing

The study uses real moving-vehicle GPS L1 C/A measurements recorded at 1575.42 MHz with a 10.23 MHz sampling rate. GNSS-SDR tracks visible satellites and decodes their navigation data. Each satellite is processed in a separate signal stream. The decoded navigation-bit signs are removed before complete 40-ms coherent observations are formed. Correlation screening identifies observations that may contain an additional signal component.

Per-satellite SAGE estimation

For satellite stream m , the received complex baseband signal is modeled as

$$r_m(t) = \sum_{\ell=0}^{L_m-1} \alpha_{m,\ell} s_m(t - \tau_{m,\ell}) e^{j2\pi\nu_{m,\ell}t} + n_m(t).$$

SAGE estimates the delay, Doppler, and complex gain of local signal components. For each 40-ms observation, candidate path counts $L = 1-4$ are compared using BIC and residual power. Delay is refined at 0.1-sample spacing within a ± 0.8 -sample neighborhood, and Doppler is searched within ± 30 Hz of the tracking estimate in 5-Hz steps.

Persistent path observations

Adjacent observations are matched using thresholds of 1.5 samples for excess delay, 40 Hz for signed relative Doppler, and 10 dB for relative power. A component is retained as persistent when matches occur in at least three consecutive observations. The reported path parameters are relative to the earliest estimated component:

$$\Delta\tau = \tau_\ell - \tau_0, \quad \nu_{\text{rel}} = \nu_\ell - \nu_0, \quad P_{\text{rel}} = 10 \log_{10} \frac{|\alpha_\ell|^2}{|\alpha_0|^2}.$$

Statistical descriptions

Each satellite track receives the same total weight so that long tracks do not dominate the environment-level summaries. Excess delay is represented by a shifted lognormal model in Urban and a lognormal model in Mountain/Valley. Signed relative Doppler is represented by a track-balanced empirical CDF. Relative power is represented by a two-component Gaussian mixture. The power–delay trend is represented by two connected line segments with a common breakpoint.

2. Core results

Analyzed observations

The analysis contains 518 persistent MPC observations from 236 satellite tracks, 36 recordings, and 9 measurement scenes. Urban contributes 346 observations and Mountain/Valley contributes 172 observations.

Excess delay

Both environments show right-skewed excess-delay distributions. The Urban shifted-lognormal fit has a shift of 68.7 ns, a scale of 102.8 ns, a log-domain standard deviation of 0.675, and a maximum weighted CDF deviation $D_w = 0.082$. The Mountain/Valley lognormal fit has a scale of 187.8 ns, a log-domain standard deviation of 0.348, and $D_w = 0.055$.

Signed relative Doppler

The track-balanced empirical-CDF quantiles are:

Environment	P_{10}	P_{50}	P_{90}
Urban	−55.6 Hz	19.5 Hz	80.6 Hz
Mountain/Valley	−94.4 Hz	−45.1 Hz	55.3 Hz

Mountain/Valley has a more negative P_{10} and median, whereas the Urban median remains positive.

Relative power

Both environments show two relative-power modes near −13 dB and −3 to −4 dB relative to the earliest component. The two-component GMM gives $D_w = 0.045$ in Urban and $D_w = 0.037$ in Mountain/Valley, compared with 0.133 and 0.137 for a single Gaussian.

Power–delay relation

The connected two-slope model has a common breakpoint at 190.6 ns. The Urban slopes are −3.53 and −5.57 dB per 100 ns before and after the breakpoint, with RMSE 4.01 dB. The Mountain/Valley slopes are −6.31 and −4.46 dB per 100 ns, with RMSE 3.44 dB.

Use of the results

The fitted delay and relative-power marginals, the empirical Doppler CDF, and the connected power–delay trend provide path-level inputs for simplified vehicular satellite-to-ground simulation. They also provide empirical propagation evidence relevant to GNSS and NTN-assisted localization.